1. (a) True. That common number of elements is what we call the dimension.

(b) False. What is described is equivalent to linear dependence.

(c) False. One reason is that $\vec{0}$ is not in the set.

(d) False. $\mathbb{R}^n = \{\vec{0}\}$, the trivial subspace, which contains the zero vector.

(e) False. $\vec{b} \in Col(A)$ makes the problem consistent, but there may be many solns.

(f) True. $\|\vec{x}+\vec{y}\|^2 = \langle \vec{x}+\vec{y} \mid \vec{x}+\vec{y} \rangle = \langle \vec{x} \mid \vec{x} \rangle + \langle \vec{x} \mid \vec{y} \rangle + \langle \vec{y} \mid \vec{x} \rangle + \langle \vec{y} \mid \vec{y} \rangle$

$$= \|\vec{x}\|^2 + 2\langle \vec{x} \mid \vec{y} \rangle + \|\vec{y}\|^2.$$

If $\|\vec{x}+\vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2$, this means $\langle \vec{x} \mid \vec{y} \rangle = 0$. And, if $\langle \vec{x} \mid \vec{y} \rangle = 0$, then $\|\vec{x}+\vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2$.

2. There are many examples, such as

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

3. (a) We are given that $[A \mid \vec{0}] \longrightarrow [U \mid \vec{0}]$. This means there are 3 free variables, x_2, x_4 and x_5 and the equations say

$$\left.\begin{aligned} 2x_1 + x_2 + x_4 &= 0 \\ -x_3 - 3x_4 + 5x_5 &= 0 \end{aligned}\right\} \Rightarrow \begin{aligned} x_1 &= -\tfrac{1}{2}x_2 - \tfrac{1}{2}x_4 \\ x_3 &= -3x_4 + 5x_5 \end{aligned}$$

Take $\underbrace{x_2 = 2,\ x_4 = 0,\ x_5 = 0}$; take $\underbrace{x_2 = 0,\ x_4 = 2,\ x_5 = 0}$; take $\underbrace{x_2 = 0,\ x_4 = 0,\ x_5 = 1}$

$$\Downarrow \qquad\qquad \Downarrow \qquad\qquad \Downarrow$$

$$\begin{bmatrix} -1 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} -1 \\ 0 \\ -6 \\ 2 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \\ 1 \end{bmatrix}$$

These 3 vectors comprise a basis of $Null(A)$.

(b) $(Null(A))^{\perp} = Row(A)$, whose dimension is $rank(A) = 2$.

(c) Let $\vec{y} = U\vec{x}$, so $LU\vec{x} = \vec{b}$ becomes $L\vec{y} = \vec{b}$. Using backward substitution

$$\left.\begin{aligned} y_1 &= 3 \\ 2y_1 + y_2 &= 8 \\ -y_1 + 3y_2 + y_3 &= 3 \\ 3y_1 + y_3 + y_4 &= 9 \end{aligned}\right\} \Rightarrow \vec{y} = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

Now we solve $U\vec{x} = \vec{y}$

$$\begin{aligned} 2x_1 + x_2 + x_4 &= 3 \\ -x_3 - 3x_4 + 5x_5 &= 2 \end{aligned}$$

We get a particular soln. setting the 3 free vars. $x_2 = x_4 = x_5 = 0$: $\vec{x}_p = \begin{bmatrix} -3/2 \\ 0 \\ -2 \\ 0 \\ 0 \end{bmatrix}$

The full soln. comes from combining this with part (a):

$$\begin{bmatrix} -3/2 \\ 0 \\ -2 \\ 0 \\ 0 \end{bmatrix} + r \begin{bmatrix} -1 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ -6 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \\ 1 \end{bmatrix}, \quad r, s, t \in \mathbb{R}.$$

4. By introducing $s_1 = x_1 + x_2 - 4$, the inequality

$x_1 + x_2 \geq 4$ becomes $\underline{x_1 + x_2 - s_1 = 4}$ and $s_1 \geq 0$.

Similarly, the slack variables $s_2 = x_1 + 3x_2 - 12$ and $s_3 = x_1 - x_2$,

$x_1 + 3x_2 \geq 12$ becomes $\underline{x_1 + 3x_2 - s_2 = 12}$ and $s_2 \geq 0$

$x_1 - x_2 \geq 0$ becomes $\underline{x_1 - x_2 - s_3 = 0}$ and $s_3 \geq 0$.

Putting all the variables into $\vec{x} = [x_1 \ x_2 \ s_1 \ s_2 \ s_3]^T$, our linear programming problem becomes

$$\min \underbrace{\begin{bmatrix} 2 & 1 & 0 & 0 & 0 \end{bmatrix}}_{c^T} \vec{x} \quad \text{subject to} \quad \underbrace{\begin{bmatrix} 1 & 1 & -1 & 0 & 0 \\ 1 & 3 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 & -1 \end{bmatrix}}_{A} \vec{x} = \underbrace{\begin{bmatrix} 4 \\ 12 \\ 0 \end{bmatrix}}_{\vec{b}}$$

5. (a) Basic variables correspond to pivot columns: x_3, x_4, x_5

(b) Free variables $x_1 = x_2 = 0$, so initial position is $(0, 0, 4, 12, 18)$

(c) Cost $= -3x_1 - 5x_2$, showing that if x_2 is made positive, cost will decrease most quickly. So, x_2 will migrate from free to basic. Correspondingly, the simplex method algorithm calls for x_4 to migrate from basic to free, since the ratios to consider are $\frac{4}{0}$ (like $+\infty$), $\underline{\frac{12}{2} = 6}$, and $\frac{18}{2} = 9$

smallest, and corresponds to the row in which x_4 is the pivot var.